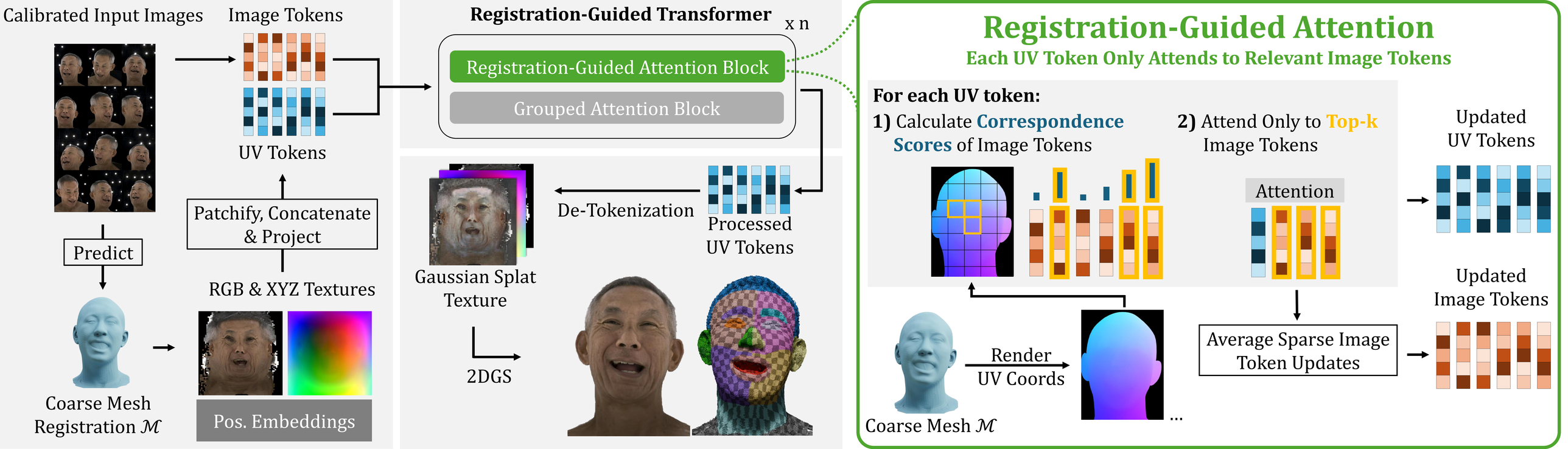
Calibrated Input Images
Image Tokens
UV Tokens
Patchify, Concatenate & Project
Predict
RGB & XYZ Textures
Coarse Mesh Registration $\mathcal{M}$
Pos. Embeddings
Registration-Guided Transformer
x n
Registration-Guided Attention Block
Grouped Attention Block
Gaussian Splat Texture
De-Tokenization
Processed UV Tokens
2DGS
Registration-Guided Attention
Each UV Token Only Attends to Relevant Image Tokens
For each UV token:
1) Calculate Correspondence Scores of Image Tokens
2) Attend Only to Top-k Image Tokens
Attention
Updated UV Tokens
Updated Image Tokens
Average Sparse Image Token Updates
Render UV Coords
Coarse Mesh $\mathcal{M}$
...